

Software Testing Scope & Strategy Document

E-Commerce Quality Assurance Portfolio Project

Target System: <https://sauce-demo.myshopify.com/>

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1. Document Overview

This document outlines the testing scope, key functional modules, and coverage strategy for the **SauceDemo E-Commerce** web application. The objective is to establish clear boundaries for manual test case execution and identify high-priority flows for automated UI testing.

2. In-Scope Functional Areas

2.1 Product Browsing & Selection

Focuses on user discovery and product detail interaction prior to purchase.

- **Product Catalog View:** Verifying product card rendering, images, titles, and price labels.
- **Variant Selection:** Validating dynamic price and image updates when toggling options (e.g., Size, Color).
- **Search & Filtering:** Testing keyword search functionality and catalog filter behavior.

2.2 Shopping Cart Management

Focuses on core cart operations and financial subtotal accuracy.

- **Item Addition:** Verifying single and multi-quantity additions from Product Detail Pages (PDP).
- **Cart Updates:** Adjusting line item quantities dynamically and asserting subtotal recalculation.
- **Item Removal:** Deleting items from the cart and confirming empty cart state behaviors.
- **State Persistence:** Ensuring cart contents persist across session refreshes and navigation.

2.3 Checkout Flow & Field Validations

Focuses on user data entry integrity and navigation leading to order completion.

- **Guest Checkout Navigation:** Transitioning smoothly from cart drawer to checkout pages.
- **Form Field Validation:** Testing boundary conditions, mandatory field checks, and inline error handling (e.g., malformed email addresses, missing ZIP/postal codes).
- **Discount & Shipping Options:** Verifying promotion code entry evaluation and shipping method selections.

2.4 User Interface & Usability Features

Focuses on cross-device presentation and general UI health.

- **Responsive Design:** Validating UI layout adaptation across Desktop, Tablet, and Mobile viewports.
- **Promotional Elements:** Testing banner links, modal popups, and footer navigation.

3. Automation vs. Manual Testing Breakdown

Feature Area	Testing Strategy	Automation Priority
Product Browsing	Manual UI checks + Search API/UI automation	Medium
Shopping Cart	Full End-to-End UI Automation (Selenium/Playwright)	High
Checkout Validations	Negative input validation via UI Automation	High
UI / Responsiveness	Cross-browser visual manual checks	Low

Table 1: Test Coverage Allocation

4. Out of Scope

- Real payment gateway processing (credit card transactions on live merchant networks).
- Backend database server performance and load stress testing.